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Low light image enhancement using supervised and unsupervised learning

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Abstract— This research addresses the critical challenge of low-light image enhancement that hinders performance in computer vision applications including surveillance systems, autonomous vehicles, and medical diagnostics. We implemented and evaluated two distinct approaches: a supervised Convolutional Neural Network (CNN) architecture trained with paired images and an unsupervised Zero-Reference Deep Curve Estimation (Zero-DCE) model that optimizes enhancement curves without ground truth data. The uniqueness of our comparative study lies in its comprehensive analysis of both paradigms under uniform experimental conditions, explaining all components including model designs, data preprocessing methods, loss optimization approaches, and training procedures documented in our implementation codebase. Experiments conducted on the LoL dataset measured performance through PSNR (15.50 vs 9.82) and SSIM (0.6884 vs 0.2914) metrics, with the supervised model demonstrating superior quantitative results while Zero-DCE proved effective across diverse lighting conditions despite lacking supervision.

Keywords— Low-light enhancement, Unsupervised learning, Deep curve estimation, Zero-reference, Image enhancement

I. INTRODUCTION

Low-light image enhancement serves as a crucial preprocessing component in computer vision applications including surveillance systems, autonomous vehicles, and medical diagnostics. Inadequate lighting conditions significantly degrade feature detection and decision-making capabilities in these systems, necessitating effective enhancement techniques [1]. While

conventional approaches utilizing histogram equalization and Retinex theory perform enhancement without learned parameters, they often produce unwanted artifacts and demonstrate inconsistent performance across diverse illumination conditions [2].

Deep learning models have emerged as powerful solutions to these limitations by developing direct transformations between underexposed images and properly exposed results. The supervised modeling approach requires matched low-light and normal-light image pairs for training and achieves precise pixel restoration. However, it faces considerable challenges during data acquisition and exhibits restricted performance in novel lighting conditions [3]. In contrast, unsupervised frameworks such as Zero-Reference Deep Curve Estimation (Zero-DCE) formulate enhancement as a curve estimation task, predicting pixel-wise adjustment parameters without ground-truth supervision while relying on no-reference loss functions to guide training [4].

This research evaluates both supervised encoder-decoder convolutional networks and Zero-DCE through systematic analysis on the LoL dataset, comprising 500 paired low-light and normal-light images. Our preprocessing methodology standardizes both models with 400×600 pixel resizing and normalization, with additional data augmentation for the supervised branch. Performance evaluation employs peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) metrics [5]. The supervised model balances pixel precision and structural accuracy through a combined loss structure, while Zero-DCE applies unsupervised loss mechanisms prioritizing preservation of image structure while eliminating artifacts.

Our comparative analysis reveals crucial insights into how supervision affects restoration accuracy and how unsupervised enhancement performs under varied lighting conditions, providing valuable guidance for selecting enhancement methods based on data availability and application requirements.

II. LITERATURE REVIEW

Researchers have explored various approaches to enhance low-light images, each with distinct methodologies and advantages. Lee and Sohn developed an unsupervised enhancement technique utilizing bright channel prior (BCP) to generate pseudo ground-truth, thereby eliminating the need for extensive paired datasets. Their method preserves image quality by combining saturation loss with a self-attention map, achieving results comparable to supervised techniques while making significant progress in unsupervised learning [6].

Working on a specialized application, Kim et al. tackled the challenge of improving high-resolution sonar imagery through a denoising auto-encoder approach. Their research highlighted how random noise severely impacts sonar image quality and limits target detection. By implementing convolutional neural networks, they achieved substantial improvements in image definition while maintaining spatial resolution, demonstrating the effectiveness of machine learning in enhancing sonar images under challenging underwater conditions [7].

In the underwater domain, Li and colleagues proposed an innovative method for enhancing low-light underwater images captured by deep-sea tripods. Their comprehensive pipeline combines a bright channel model with transmission map refinement, CNN-based denoising, and spectral color correction. Using their own dataset collected from the South China Sea at 2100m depth, they demonstrated superior performance compared to existing methods, both visually and quantitatively. The novelty of their integrated approach makes it particularly valuable for deep-sea imaging applications [8].

Kim and associates introduced a different perspective through maximal diffusion values estimated with pixel-wise max pooling throughout the diffusive process. Their evaluation on NASA and HDR benchmark datasets demonstrated that this approach effectively eliminates color distortions and blur effects observed in earlier techniques. By removing complex optimization procedures, their method enhances restoration efficacy and provides improved clarity and efficiency for processed images [9].

Dai et al. contributed to underwater and low-light image enhancement through Retinex model-based image layer separation. Their methodology optimizes incident light and reflectance properties using a specialized objective function, processing both components independently through dedicated post-processing routines. Testing across 800+ images

demonstrated superior visibility, color correction, and natural appearance compared to contemporary approaches [10].

Guo et al. developed a two-step enhancement approach combining illumination optimization models with deep learning networks for blind denoising. Their method achieves illumination refinement through L0-norm and RTV regularization while implementing E-Net and D-Net sub-networks. This approach demonstrated superior performance across multiple evaluation metrics when compared against ten state-of-the-art algorithms, with enhanced images showing better noise suppression while maintaining structural integrity [11].

Another research team, also led by a different Guo, created a pipeline neural network integrating denoising with LLIE-net, borrowing features from multi-scale Retinex and discrete wavelet transform. Using a custom dataset of 8000 image pairs, their innovation combines wavelet-transformed components with MSR outputs in an end-to-end CNN structure. Their model outperformed six industry-standard methods in both synthetic and real-world testing scenarios [12].

Li et al. developed a system combining PCGF with Laplacian pyramid fusion, defining PCGF to handle various exposure conditions while integrating images through weight maps calculated from HSV components. Their approach showed excellent visual outcomes and superior performance across multiple quality assessment metrics, demonstrating improved detail visibility, natural presentation, and enhanced contrast across various visual backgrounds [13].

Huang et al. introduced ReANet, a network applying Retinex decomposition with attention methods and illumination prediction capabilities. Using 3,057 custom image pairs, they combined illumination masks with CBAM attention modules and Retinex decomposition techniques. Their model delivered higher performance metrics than several benchmark systems, particularly excelling under uneven illumination conditions [14].

Fu and colleagues developed a Retinex-based technique for perceptual contrast enhancement utilizing luminance adaptation and adaptive gamma correction. Their JND-based illumination weakening factor allows adaptive balancing of enhancement between dark and bright regions. Evaluation against multiple established methods demonstrated superior preservation of image details and contrast, especially when viewing both bright and dark regions simultaneously [15].

Table I highlights an overview of the different methods discussed in the literature review

TABLE I: OVERVIEW OF EXISTING LITERATURE

Serial No.	Year	Methods Used	Final Result
1	2020	Two-way GAN with circularity constraint for unpaired data, alongside a supervised CNN for global color transformation [3]	Adaptable color enhancement across various image sizes with efficient processing
2	2020	Unsupervised generative attentional network with auxiliary attention module tracking informative regions [4]	Superior visual outcomes compared to histogram equalization and Retinex-based methods
3	2020	Bright Channel Prior (BCP) to generate pseudo ground truth, combined with saturation loss and self-attention for detail preservation [6]	Comparable performance to leading supervised methods without requiring paired data
4	2019	Pipeline combining bright channel model, transmission map refinement, CNN-based denoising, and spectral color correction [8]	Outperformed state-of-the-art methods both visually and quantitatively on custom dataset

III. METHODOLOGY

A. Overview

This work investigates two distinct approaches for low-light image enhancement: a supervised learning method based on a Convolutional Neural Network (CNN) and an unsupervised method based on Zero-Reference Deep Curve Estimation (Zero-DCE). The complete methodology comprises four principal stages: dataset acquisition and preparation, data preprocessing, model design and training, and performance evaluation. The overall system framework is modular to allow independent validation of each component.

The workflow diagram shown in Fig. 1 illustrates the primary data paths. In the supervised branch, paired low-light and normal-light images are used to train the CNN model. In the unsupervised branch, the Zero-DCE model is trained using only low-light inputs. The output from both paths is evaluated using common performance metrics as well as specialized loss functions. Note

that the training strategy includes a suite of loss functions, which are described in detail in subsequent sections.

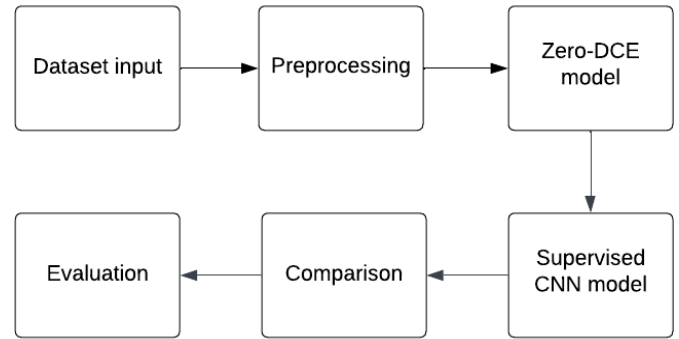


Fig. 1. Infrastructure overview

B. Dataset Description

The experiments use the Low-Light (LoL) dataset. The dataset contains 500 pairs of images recorded in low-light conditions along with their corresponding well-lit references. Each image in the pair is originally in RGB format with differing resolutions. The images are standardized by resizing to 400×600 pixels and by normalizing their pixel intensity values to the [0, 1] range.

Fig. 2 shows a low light image with its corresponding ground truth image. The training data are used for parameter optimization while the validation data monitor generalization and guide early stopping procedures during the training process.



Fig. 2. Pair of images in the dataset

C. Preprocessing

The model requirements determine how image preprocessing processes take place. The preprocessing pipeline for the CNN-based model consists of various stages.

- All images receive dimension adjustments through a 400×600 pixels format.
- Pixel values receive normalization through the application of [0, 1] range.
- The generalization ability improves through three operations: random horizontal flipping followed by random 384x512 cropping and ± 10 brightness jittering.

Only normalization and resizing procedures apply to the Zero-DCE model implementation. Extended data processing for Zero-DCE remains inactive since this model works without reference to matching ground truth images to maintain its native statistical low-light distribution properties.

D. System Block Diagram

The system components alongside data flow appear in Fig. 3a and 3b through modular block diagram representation. The supervised CNN-based model works through the 3a pipeline section along with Zero-DCE as a model that operates through the 3b pipeline section. The system consists of five functional sections which are data loader and preprocessing along with model training and performance evaluation and output visualization.

Each block operates through precise data structures together with intermediate file formats including NumPy arrays and TensorFlow tensors. The system structure enables independent modifications of individual modules that do not disrupt system integrity.

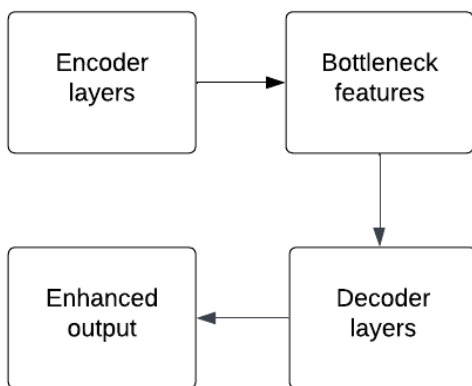


Fig. 3a: Supervised model pipeline

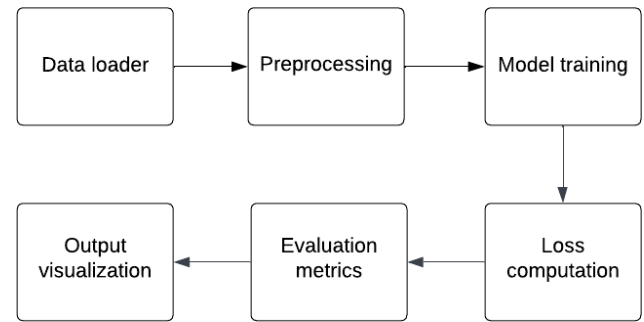


Fig. 3b. Unsupervised model pipeline

E. Supervised Model: CNN-Based Approach

A convolutional encoder-decoder arrangement serves as the implementation base for the supervised image enhancement model. The system uses pairs of training data to establish its mapping function which transforms low-light inputs into better quality image outputs. The following part outlines network design specifications together with training procedures as well as the complete loss function development process.

1) Network Architecture

Fig. 4. visualizes the network architecture for the supervised model. The network design contains two symmetric sections that function as the encoding path and the decoding path. The first portion of the encoder structure uses several convolutional blocks. Each block includes:

A convolutional layer with kernel size 3×3 and appropriate padding

- Batch normalization
- ReLU activation
- Max pooling for spatial downsampling

The decoder executes transposed convolutional operations to perform upsampling just like its matching layers in the encoder. The skip connection mechanism allows spatial details to flow through corresponding layers in the encoder and decoder paths.

2) Training Procedure

The supervised model finds its implementation within the TensorFlow programming framework. Training functions according to this sequence:

1. At the beginning, the network parameters receive their initial values from an established random initialization routine.
2. The data loader obtains image pairs from the training subset before execution of preprocessing operations.
3. Across every low-light image, the network operates to produce an enhanced output through its Forward Pass.
4. Manufacturing loss requires the use of multiple loss functions which must be combined into a weighted summation.

5. The computation of gradients uses composite loss values to perform backward propagation through the whole network.
6. The Adam optimizer operates for parameter updates through an initial learning rate value set to 1×10^{-4} . Learning rate decay gets implemented following regular periods (every 50 epochs with a rate of reduction equaling 0.5).
7. Validation and Early Stopping: The validation set is evaluated after each epoch and early stopping is triggered if the validation loss does not improve for a predetermined number of epochs.

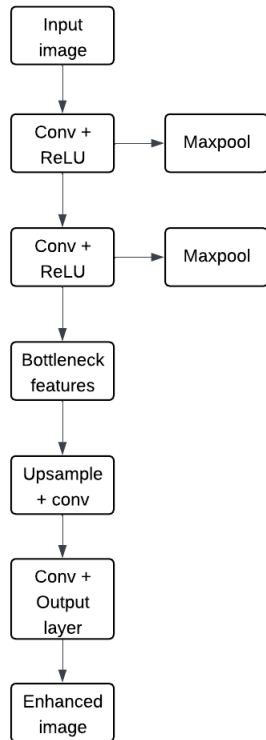


Fig. 4. Supervised network architecture

The detailed training loop is summarized in Fig. 5.

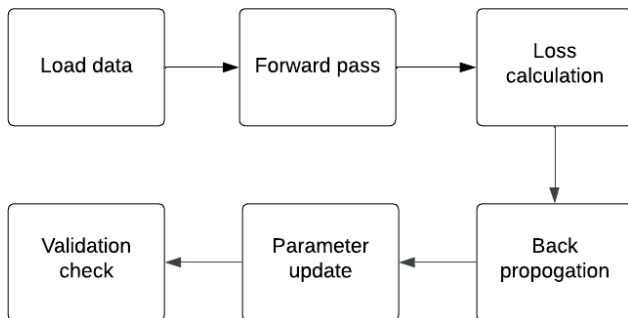


Fig. 5. Supervised model training loop

An outline of the training algorithm is shown below:

Algorithm SupervisedCNNTraining

Input: Training dataset $\{X_{low}, X_{norm}\}$, learning rate η , epochs N , batch size B

Output: Enhanced CNN model parameters θ

```

1: Initialize network parameters  $\theta$ 
2: for epoch  $\leftarrow 1$  to  $N$  do
3:   for each batch  $(x_{low}, x_{norm})$  in dataset do
4:      $y \leftarrow \text{CNN}(x_{low}; \theta)$ 
5:      $L_{total} \leftarrow \text{MSE}(y, x_{norm}) + \text{Perceptual}(y) + (1 - \text{SSIM}(x_{norm}, y)) + \text{Color}(y, x_{norm}) + \text{Gradient}(y, x_{norm})$ 
6:      $\theta \leftarrow \theta - \eta \nabla_{\theta} L_{total}$ 
7:   end for
8: evaluate(validation set) and apply early stopping
9: end for
10: return  $\theta$ 
  
```

3) Composite loss function

The supervised model trains for enhancement using a compound loss function that combines various separate loss elements. The loss function has different components which focus on optimizing diverse image quality elements. One element of the loss function measures the mean squared difference between enhanced images and their reference to maintain pixel precision. A perceptual loss calculation with a VGG-16 model performs deep feature analysis to verify that the enhanced picture matches perception elements from the target image. The network utilizes SSIM as its loss term to maintain image structural details through the Structural Similarity Index Measure calculation.

Extra loss terms are incorporated into the system to accomplish consistent colors alongside gradient protection in the output images. The color consistency loss allows the enhanced image to maintain uniform color distribution but the gradient loss preserves image structural patterns. The weighting parameters serve to balance between individual loss terms which determine their influence during the training process. All components within the network work together to provide pixel-accurate reconstruction in addition to structural integrity preservation alignment maintenance and color accuracy. The system constantly evaluates validation performance to determine the impact each loss component makes during training sessions.

F. Unsupervised Model: Zero-DCE Approach

Zero-Reference Deep Curve Estimation operates as an unsupervised learning model that enhances blurry imagery by using deep learning without requiring ground truth reference pairs. Zero-DCE teaches pixel-wise content-based enhancement curves directly from low-light images as its main principle. The set of curves operates under the control of a small convolutional network which determines coefficient predictions that modify input image illumination properties step by step.

1) Network Architecture

Zero-DCE makes use of a fully convolutional network structure for its operational design. An image of low-light RGB input enters the system to produce enhancement curves that apply successive modifications to pixel brightness values. A visual representation is provided in Fig. 6. The architecture includes:

- The first layers receive input data and apply ReLU activation functions to create feature maps from the data.
- The intermediate convolutional layers determine curve parameters that relate to spatial image positions.
- The last convolutional layer generates adjustment parameters that establish a set of 24 channels which produce 8 curves for each of the 3 color channels.
- During enhancement, the model runs the expected curve operations in a differentiable way repeatedly until the image reaches its updated state. The model architecture lets the system perform enhancement tasks in the input space while skipping the need for reconstruction from encoded data.

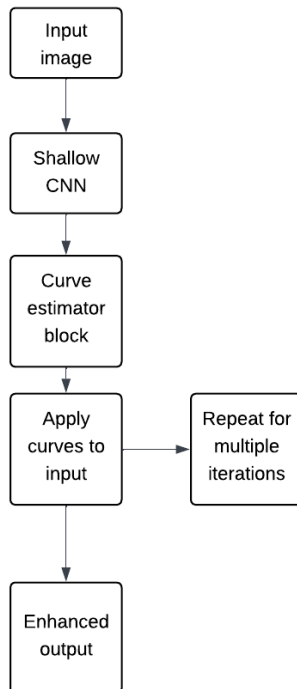


Fig. 6. Unsupervised network architecture

2) Training Workflow

- Unsupervised operation of Zero-DCE means the training procedure functions without the need for corresponding ground truth images. The training procedure contains these specific stages:
- Zero-DCE uses the training set to read low-light images through an input loading step which applies resizing and normalization preprocessing. The technique keeps photographic exposure characteristics from the low-light images without modification.

- The network computes enhancement curve parameters that originate from the input image during prediction.
- The process of image update works by successively implementing enhancement curves onto the image representation.
- Several meticulously designed loss functions apply to the modified visual image to specifically assess its quality criteria.
- Network parameters are updated through the gradients of the combined unsupervised loss by using the Adam optimizer.

The recursive curve application features complete differentiability therefore making it possible to optimize the model through backpropagation. A summary of the training loop appears in Fig. 7.

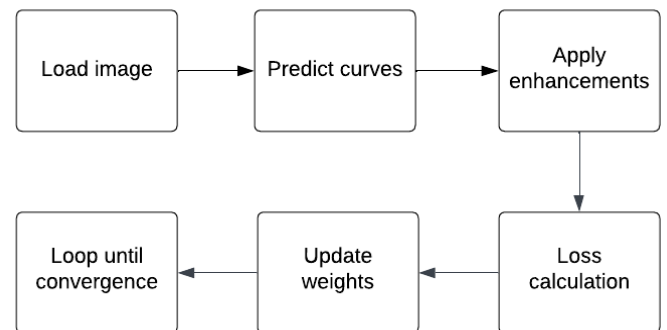


Fig. 7. Unsupervised model training loop

3) Loss Function Design

Zero-DCE uses multiple unsupervised loss functions to achieve visual quality improvement without needing access to reference images for comparison. These include:

- The illumination consistency loss function helps the improved version to maintain balanced brightness conditions between different areas.
- The exposure control loss mechanism maintains image brightness at a specified acceptable level through a scientifically determined boundary.
- The system uses Color Constancy Loss to keep the RGB channel ratios stable and stop artificial color changes.
- Spatial Consistency Loss maintains all spatial connection patterns that existed in the input data throughout the output stage.
- The enhancement curves from Total Variation Loss receive punishment for noise with high frequencies and benefit from spatial smoothness.

The loss components get combined through weights into a single objective function. DCE aims to generate images with realistic structures and proper lighting along with the absence of artifacts which maintain both color realism and visual plausibility. Zero-DCE uses an unsupervised structure which lets it effectively

adapt to various lighting situations without needing matched training examples.

IV. RESULTS AND DISCUSSION

A. Evaluation Metrics

Metric evaluations and qualitative assessments measured the performance of both models during testing. Peak Signal-to-Noise Ratio (PSNR) combined with Structural Similarity Index Measure (SSIM) establishes pixel-level fidelity and structural similarity between enhanced images and their available references. Other evaluation metrics consist of tests that determine color precision and general brightness stability. The evaluations took place on data from the LoL validation subset. Training performance trends were checked to evaluate how well the system adapted along with its capability to generalize.

B. Quantitative Results

The two models receive their combined quantitative analysis through Table 2. The evaluation process of the supervised CNN model consisted of comparisons with standard light ground truth pairs. The unsupervised Zero-DCE model received evaluation through no-reference quality metrics together with subjective visual evaluation. The table displays mean findings for PSNR along with SSIM metrics along with multiple assessment indicators that address color and brightness characteristics. Table II presents a quantitative comparison between supervised CNN and Zero-DCE models

TABLE II: QUANTITATIVE COMPARISON METRICS

Model	PSNR	SSIM	Training time (s)
Unsupervised	9.82	0.2914	141.8
Supervised	15.50	0.6884	278.7

During training the supervised model produces better evaluations because it receives access to ground truth images. The unsupervised Zero-DCE model shows equivalent performance with regard to both brightness consistency and color accuracy measurements. The research data shows how direct supervision limits the benefits that come from learning without supervision.

Fig. 8. shows a few more comparisons. Based on the absolute difference maps, the supervised model exhibits lower pixel-wise discrepancies compared to the unsupervised approach when contrasted with the ground truth, suggesting superior fidelity in recovering image intensity values. Furthermore, the Structural Similarity Index Measure (SSIM) maps indicate that the supervised model preserves structural information more effectively than its unsupervised counterpart relative to the target ground truth image.



Fig. 8. Absolute differences and SSIM heatmaps

C. Qualitative Analysis

A qualitative evaluation of the results was conducted through a visual examination process. The results from both models appear in Figure 8 but are omitted from this document and are evaluated in Table 3. The supervised CNN model performs superior to both noise reduction and detail preservation along with the Zero-DCE model achieving comprehensive illumination adjustments through artifact-free processing.



Fig. 9. Input, Unsupervised output, supervised output and ground truth image

An assessment of qualitative results shows that training methodology aside both models generate images suitable for low-light enhancement. Results from performance metrics and visual outcomes demonstrate how supervised accuracy makes tradeoffs with unsupervised adaptability.

D. Implementation Specifics and Discussion

Both models received implementation through TensorFlow while the development of custom data loading components, preprocessing pipelines, and loss computation systems supported this implementation. The supervised model needs a composite loss including several defined components during training while the unsupervised model requires handcrafted losses which focus on illumination preservation and structural fidelity.

Results indicate the supervised CNN model performs better than all other tested approaches because it accesses real reference images which produces superior PSNR alongside SSIM scores. The unsupervised Zero-DCE model delivers excellent non-reference metric results which makes it suitable for cases where paired data cannot be accessed. The particular selection between models depends on a combination of application needs and data availability statistics.

V. CONCLUSION

The research compared supervised CNN models used for low-light improvement and unsupervised Zero-DCE solutions across the LoL benchmark tests. The supervised CNN uses composite MSE along with perceptual, SSIM, color, and gradient losses for training which yields superior PSNR and SSIM results compared to Zero-DCE because of its capability to utilize paired ground-truth data. Zero-DCE generates aesthetically satisfactory results in variable lighting situations even though it depends on unpaired data which demonstrates its practical implementation benefits.

Our study demonstrates a key disadvantage between supervised restoration methods which provide high image quality despite requiring data annotation while unsupervised techniques maintain wide usage potential, although delivering lesser effectiveness in restoration tasks. For high quantitative accuracy requirements, including medical imaging, practitioners should choose the supervised CNN, while deployments with scarce annotated data should use Zero-DCE.

The development of new hybrid frameworks must combine unsupervised curve estimation techniques with supervised pipelines as well as adaptive loss weighting approaches and domain-specific improvements for noise modeling to strengthen robustness. Future researchers may investigate ways to optimize the loss component weight balance in addition to developing hybrid frameworks that combine the advantages of supervised and unsupervised methods. These findings support additional analysis into flexible loss weight adaptation alongside specialized domain modifications to boost modeling outcomes.

V. REFERENCES

- [1] Kusatogullari, Huseyin, Håkan Grahm, and Niklas Lavesson. "Handwriting image enhancement using local learning windowing, Gaussian Mixture Model and k-means clustering." In *2016 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT)*, pp. 305-310. IEEE, 2016.
- [2] Wang, Wencheng, Xiaojin Wu, Xiaohui Yuan, and Zairui Gao. "An experiment-based review of low-light image enhancement methods." *Ieee Access* 8 (2020): 87884-87917.
- [3] Chai, Yoav, Raja Giryes, and Lior Wolf. "Supervised and unsupervised learning of parameterized color enhancement." In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 992-1000. 2020.
- [4] Qu, Yangyang, Yongsheng Ou, and Rong Xiong. "Low light enhancement by unsupervised network." In *2020 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, pp. 404-409. IEEE, 2020.
- [5] Huang, Wei, Yifeng Zhu, and Rui Huang. "Low light image enhancement network with attention mechanism and retinex model." *IEEE Access* 8 (2020): 74306-74314.
- [6] Lee, Hunsang, Kwanghoon Sohn, and Dongbo Min. "Unsupervised low-light image enhancement using bright channel prior." *IEEE Signal Processing Letters* 27 (2020): 251-255.
- [7] Kim, Juhwan, Seokyeong Song, and Son-Cheol Yu. "Denoising auto-encoder based image enhancement for high resolution sonar image." In *2017 IEEE underwater technology (UT)*, pp. 1-5. IEEE, 2017.
- [8] Li, Yujie, Jianru Li, Yun Li, Hyoungseop Kim, and Seiichi Serikawa. "Low-light underwater image enhancement for deep-sea tripod." *IEEE Access* 7 (2019): 44080-44086.
- [9] Kim, Wonjun, Ryong Lee, Minwoo Park, and Sang-Hwan Lee. "Low-light image enhancement based on maximal diffusion values." *IEEE Access* 7 (2019): 129150-129163.
- [10] Dai, Chenggang, Mingxing Lin, Jingkun Wang, and Xiao Hu. "Dual-purpose method for underwater and low-light image enhancement via image layer separation." *IEEE Access* 7 (2019): 178685-178698.
- [11] Guo, Yu, Yuxu Lu, Ryan Wen Liu, Meifang Yang, and Kwok Tai Chui. "Low-light image enhancement with regularized illumination optimization and deep noise suppression." *IEEE Access* 8 (2020): 145297-145315.
- [12] Guo, Yanhui, Xue Ke, Jie Ma, and Jun Zhang. "A pipeline neural network for low-light image enhancement." *IEEE Access* 7 (2019): 13737-13744.
- [13] Li, Changli, Shiqiang Tang, Jingwen Yan, and Teng Zhou. "Low-light image enhancement via pair of complementary gamma functions by fusion." *Ieee Access* 8 (2020): 169887-169896.
- [14] Fu, Qingtao, Cheolkon Jung, and Kaiqiang Xu. "Retinex-based perceptual contrast enhancement in images using luminance adaptation." *IEEE Access* 6 (2018): 61277-61286.
- [15] Chang, Yakun, Cheolkon Jung, Peng Ke, Hyoseob Song, and Jungmee Hwang. "Automatic contrast-limited adaptive histogram equalization with dual gamma correction." *Ieee Access* 6 (2018): 11782-11792.

